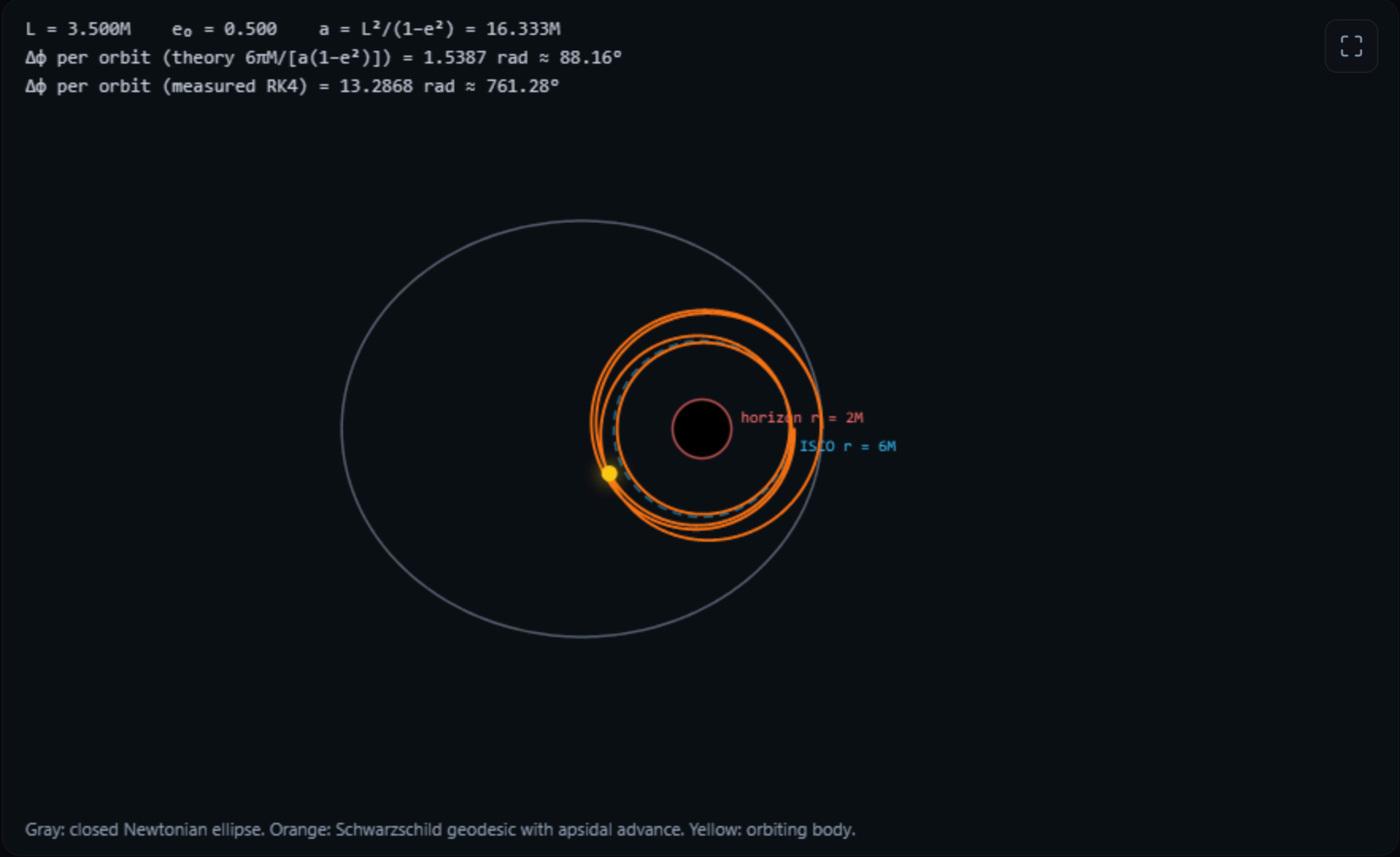


Schwarzschild Orbit Precession (Rosette)

Schwarzschild geodesic in the ϕ -form $d^2u/d\phi^2 + u = 1/L^2 + 3u^2$ ($G = c = M = 1$) integrated by RK4. The closed Newtonian ellipse is replaced by an orange precessing rosette with apsidal advance $\Delta\phi \approx 6\pi M/[a(1 - e^2)]$ per orbit — the same mechanism that produces the historic 43"/century perihelion shift of Mercury. Horizon $r = 2M$ and ISCO $r = 6M$...



GEODESIC

Angular momentum L (M) 3.5

Eccentricity at periapsis 0.5

Orbits 4

- ☒ Show Newtonian ellipse
- ☒ Show horizon r = 2M
- ☒ Show ISCO r = 6M
- ☒ Animate body

Restart body

In Schwarzschild geometry the orbit equation gains a $3u^2$ term: $d^2u/d\phi^2 + u = 1/L^2 + 3u^2$ with $u = 1/r$, $c = G = M = 1$. The closed Newtonian ellipse is replaced by a precessing rosette with apsidal advance $\Delta\phi \approx 6\pi M/[a(1-e^2)]$ per orbit — the textbook GR test that delivers the historic 43"/century perihelion advance of Mercury. Compare the gray Newton ellipse and the orange GR rosette; both share L and the initial periapsis, isolating the GR effect.

MEASURED VALUES

a (M)	16.3333
$\Delta\phi$ theory	88.163°
$\Delta\phi$ RK4	761.278°

About this model

Test-particle motion in the Schwarzschild metric is integrated in the ϕ -form $d^2u/d\phi^2 + u = 1/L^2 + 3u^2$ with $G = c = M = 1$ and $u = 1/r$, using RK4. The Newtonian closed ellipse becomes an orange precessing rosette; the apsidal advance per radial period is $\Delta\phi \approx 6\pi M/[a(1 - e^2)]$, the same GR effect behind Mercury’s historic 43"/century perihelion shift. Horizon $r = 2M$ and ISCO $r = 6M$ are annotated. Assumptions: geodesic motion of a test mass, Schwarzschild (non-spinning) spacetime, and geometric units—no radiation reaction. Vary orbital parameters to compare nearly Newtonian ellipses with strongly precessing rosettes near the hole.

Who it's for: Advanced undergraduate and graduate general relativity and relativistic astrophysics courses.

KEY TERMS

- Schwarzschild geodesic
- perihelion precession
- apsidal advance
- rosette orbit
- ISCO
- event horizon

How it works

Schwarzschild orbit equation $d^2u/d\phi^2 + u = 1/L^2 + 3u^2$ integrated by RK4 ($G = c = M = 1$). The orange GR geodesic precesses by $\Delta\phi \approx 6\pi M/[a(1-e^2)]$ per orbit — the same mechanism that produces the famous 43"/century perihelion advance of Mercury. A gray Newtonian ellipse is overplotted to isolate the GR correction; ISCO ($r = 6M$) and horizon ($r = 2M$) are marked.

Frequently asked questions

Why does the ellipse fail to close?

The extra $3u^2$ term in the orbit equation is a GR correction absent from Newtonian $1/r$ potentials. It advances the periapsis each radial period so the tip of the ellipse slowly rotates, drawing a rosette. This is spacetime curvature, not a classical inverse-cube ‘extra force’ in flat space—though the math looks similar in the $u(\phi)$ equation.

How is this related to Mercury’s 43″/century?

The weak-field limit of $\Delta\phi \approx 6\pi M/[a(1 - e^2)]$ per orbit, accumulated over a century for Mercury’s a and e , yields about 43 arcseconds after other Newtonian perturbations are removed. The simulator shows the strong-field version of that same mechanism in geometric units.

What do $r = 2M$ and $r = 6M$ mark?

$r = 2M$ is the Schwarzschild event horizon; timelike circular orbits do not exist stably down to the horizon. The innermost stable circular orbit sits at $r = 6M$ for a non-spinning hole—inside that, circular orbits are unstable. The annotations orient the rosette relative to those scales.

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Mercury Perihelion Precession

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GR $\Delta\omega$ per orbit vs Newton; ~43″/century readout; amplified animation.

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Leading-order post-Newtonian inspiral of a compact binary: $f(\tau) \propto \tau^{-(3/8)}$, strain $h(t) \propto M_c^{5/3} f^{2/3} / \dots$

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Equal masses: Chenciner–Montgomery choreography in 2D (RK4, periodic orbit).

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ISCO & Photon Sphere (V_{eff})

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Schwarzschild effective potential $V_{\text{eff}}(r)$ for massive (timelike) and photon (null) test particles in geometric...

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Same prograde Δv at peri vs apo on one ellipse; higher ε when burning deep.

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Heliocentric path of Earth’s center: barycentric ellipse plus lunar epicycle (exaggerated).